EECS 3311 - Course Project Stage 2 Report

Project: Personal Finance AI Agent | **Author:** thickKar1 | **Date:** November 2026

1. Project Overview and Implementation Status

The Personal Finance AI Agent is an intelligent software assistant designed to handle budget tracking, transaction management, anomaly detection, bill monitoring, debt tracking, tax estimation, and multi-currency conversions. The architecture is driven by a 3-step ReAct (Reason + Act + Synthesize) loop in FinanceAgent.java and LLMClient.java. Rather than serving as a basic prompt wrapper, the AI evaluates user input, selects the appropriate localized tool, retrieves context from storage, and generates natural language responses. Both GUI (Swing) and CLI interfaces are fully functional.

2. Feature-to-Implementation Mapping

Feature ID & Name	Stage 1 Ref	Status	Implemented Classes	Key Methods	Design Patterns
F01: Budget Tracking	F01	Implemented	BudgetTool.java, BudgetTracker.java	execute(), checkBudgetLimit()	Strategy, Observer
F02: Transaction Entry	F02	Implemented	TransactionTool.java, DataManager.java	execute(), saveTransaction()	Strategy, Singleton
F03: Financial Summary	F03	Implemented	SummaryTool.java, DataManager.java	execute(), loadData()	Strategy, Singleton
F04: Anomaly Detection	F04	Implemented	AnomalyTool.java	execute(), analyzeSpending()	Strategy
F05: Bill Management	F05	Implemented	BillTool.java	execute(), getUpcomingBills()	Strategy
F06: Currency Conversion	F06	Implemented	CurrencyTool.java	execute(), convert()	Strategy
F07: Debt Tracking	F07	Implemented	DebtTool.java	execute(), calculateDebt()	Strategy
F08: Savings Goals	F08	Implemented	GoalTool.java	execute(), getGoalProgress()	Strategy
F09: Investment Monitor	F09	Implemented	InvestmentTool.java	execute(), getPortfolio()	Strategy
F10: Tax Estimation	F10	Implemented	TaxTool.java	execute(), estimateTax()	Strategy

3. Implemented Design Patterns

1. Strategy Pattern: Interfaced via FinanceTool.java across all 10 tools.
2. Factory Pattern: Implemented in ToolFactory.java to centralize tool instantiation (addresses Stage 1 feedback).
3. Facade Pattern: Implemented in ApplicationFacade.java providing unified CLI/GUI entry points.
4. Singleton Pattern: Implemented in DataManager.java managing JSON state access.
5. Observer Pattern: Implemented via BudgetTracker.java, BudgetObserver.java, and AlertNotifier.java.

4. Changes and Rationale

Original Design -> Implemented Change -> Reason:
• *Simple Wrapper -> ReAct Loop:* Updated FinanceAgent to analyze tools before execution to fulfill multi-step reasoning requirement.
• *Generic Factory -> Explicit ToolFactory:* Standardized naming in ToolFactory.java to fix Stage 1 grading feedback.
• *CLI Argument Parsing:* Updated ApplicationFacade.java to route --mode=cli directly and prevent HeadlessException.

5. AI Tools and Collaboration Log

Tool / Platform	LLM Model	Development Tasks	Token Cost	Money Cost
OpenAI API / ChatGPT	GPT-3.5-Turbo / GPT-4o	Architecture refactoring, ReAct logic, tool stubs	~45,000 tokens	$0.09 USD

GitHub Copilot	GPT-4o-mini	Boilerplate code generation & Swing UI components	Included in sub	$0.00 USD

6. Development and Verification Summary

Reporting Period	Date Range	Commit Count	Key Commit Activities & Hashes
Period 1	Oct 20 - Oct 26, 2026	5+ commits	Initial repo setup, .gitignore, Stage 1 design docs
Period 2	Oct 27 - Nov 02, 2026	5+ commits	Added .env.example, agent_core, ApplicationFacade, DataManager
Period 3	Nov 03 - Nov 09, 2026	5+ commits	Added 10 tool classes, LLMClient, fixed CLI routing (442306a)

7. Feature Demonstration Video Index

Feature ID & Name	Interface Demonstrated	Video Link
F01: Budget Tracking	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F02: Transaction Entry	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F03: Financial Summary	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F04: Anomaly Detection	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F05: Bill Management	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F06: Currency Conversion	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F07: Debt Tracking	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F08: Savings Goals	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F09: Investment Monitor	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing
F10: Tax Estimation	CLI & GUI Modes	https://drive.google.com/file/d/1e_SROuEhsqcdwOpftfZNulXx_KkDH23X/view?usp=sharing